

Lab 6, Act 2: Answer Key

An Interval for the Lethal Dose

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (1c, the comparison in 2b, 2d, the early-finisher question, and the reflection) get sample answers or discussion points instead. All numbers are computed in R when this file is knitted.

Part 2 labels 2a and 2b “Recommended: no AI” and 2c and the early-finisher question “AI optional/encouraged.” Students turn in their filled-in answers and the 2c figure, not code. AI use doesn't cost points; only undisclosed use does. Treat a mismatch between a label and a student's disclosure as something to talk about, not a penalty.

Students may keep or drop the binomial coefficients $\log \binom{n_i}{k_i}$, so NLL values are given both ways. The constant is $\sum \log \binom{n_i}{k_i} = 167.52$ for these data; differences between NLLs are the same either way.

Notation: $x = \log_{10}$ dose, $L = \text{LD50}$ on the $\log_{10}$ scale, $\text{LD50} = 10^L$ mg/L.

Part 1: Plan it on paper

1a.

Act 1 used the model $\log \frac{p}{1-p} = a + bx$, where p is the probability that a beetle dies at $\log_{10}$ dose x . Let L be the $\log_{10}$ dose at which $p = 0.5$. Find L in terms of a and b , and show that this model can be rewritten as $\log \frac{p}{1-p} = b(x - L)$. Then use Act 1's MLEs ($\hat{a} = -60.72$, $\hat{b} = 34.27$) to estimate L , and convert your estimate to mg/L.

$$\hat{L} \approx \text{_____} (\log_{10} \text{ dose}) \quad \text{LD50} = 10^{\hat{L}} \approx \text{_____} \text{ mg/L}$$

Answer. $p = 0.5$ means the log-odds is 0, so $a + bL = 0$ and $L = -a/b$. Then $a + bx = b(x + a/b) = b(x - L)$. $\hat{L} = 60.72/34.27 \approx 1.772$, and $\text{LD50} = 10^{1.772} \approx 59.1$ mg/L.

Why. $\log(0.5/0.5) = \log 1 = 0$. Factoring b out of $a + bx$ is the whole rewrite.

Watch for. $L = a/b$ (sign error, gives a negative log dose). Converting with e^L instead of 10^L (gives about 5.9, a warning sign since that's far below every tested dose). Students who say “the model changed” have missed that the two forms describe the same curves.

1b.

Write the negative log-likelihood, $\text{NLL}(L, b)$, for the eight doses. You can drop terms that don't depend on L or b .

Answer. With $p_i = 1 / (1 + e^{-b(x_i - L)})$,

$$\text{NLL}(L, b) = - \sum_{i=1}^8 [k_i \log p_i + (n_i - k_i) \log(1 - p_i)],$$

where k_i is the number killed and n_i the number exposed at dose i . An equivalent simplified form, with $\eta_i = b(x_i - L)$, is $\text{NLL} = - \sum [k_i \eta_i - n_i \log(1 + e^{\eta_i})]$.

Why. Binomial pmf, logged and summed over independent doses, with $\log \binom{n_i}{k_i}$ dropped because it doesn't involve L or b .

Watch for. Using a single p for all doses (forgetting that p depends on x_i). Writing a product with no log, or a sum with no negative sign. The simplified form is not expected.

1c.

In plain English (not R code), describe step by step how you would compute the likelihood profile for L and read a 95% interval off it.

Sample answer. (1) Choose a grid of candidate values of L around $\hat{L}$. (2) For each candidate L , hold it fixed and find the value of b that minimizes the NLL; record that minimum NLL. (3) Subtract the overall minimum NLL (at the MLE) from each recorded value. (4) Keep every L whose value is 1.92 or less; the smallest and largest of those are the ends of the 95% interval.

Why. The profile re-optimizes the other parameter at each value of the focal one; the slice in Act 1 didn't.

Watch for. Holding b at $\hat{b}$ instead of re-optimizing (that's the slice). Using 2 or 3.84 as the cutoff on the NLL scale (3.84 applies to the deviance, twice the NLL difference). A grid that is too coarse or too narrow to see the curve cross 1.92 on both sides.

Part 2: Fit and compare

```
x <- beetles$log10_dose
n_exposed <- beetles$beetles
killed <- beetles$killled

nll_ld50 <- function(par) {
  eta <- par[2] * (x - par[1])
  -sum(dbinom(killed, size = n_exposed, prob = plogis(eta), log = TRUE))
}
```

2a.

(Recommended: no AI.) Write an NLL function whose first argument is `par = c(L, b)`. Check it: at $L = 1.75$, $b = 30$ it should return about 32.19 if you used `dbinom()`, or about -135.33 if you coded the formula yourself and dropped the constant. Then fit it with `optim(..., hessian = TRUE)`, starting from your values in 1a. Use `solve()` on the Hessian to get the variance-covariance matrix; the square roots of its diagonal are the standard errors. Report the 95% quadratic interval, $\hat{L} \pm 1.96 \text{ SE}$.

```
fit <- optim(c(1.77, 34), nll_ld50, hessian = TRUE)
fit$par
```

```
## [1] 1.77172 34.27040
```

```
fit$value
```

```
## [1] 18.71513
```

```
vc <- solve(fit$hessian)
se_L <- sqrt(diag(vc))[1]
wald_L <- fit$par[1] + c(-1, 1) * 1.96 * se_L
se_L
```

```
## [1] 0.003858096
```

```
wald_L
```

```
## [1] 1.764158 1.779282
```

Answer. $\hat{L} = 1.7717$, $\hat{b} = 34.27$, NLL at the MLE = 18.72 with `dbinom()` (or -148.81 without the constant). $SE(\hat{L}) = 0.0039$; interval 1.7642 to 1.7793.

Why. Same fit as Act 1, just a different parameterization, so $\hat{b}$ and the NLL at the MLE match Act 1 and $\hat{L} = -\hat{a}/\hat{b}$. The correlation between $\hat{L}$ and $\hat{b}$ is only 0.15, so this parameterization behaves well.

Watch for. Forgetting the minus sign in the NLL (the optimizer then runs off to a boundary). Using `solve(-fit$hessian)`: `optim()` minimized the NLL, so no sign flip is needed; a negative variance is the giveaway. Rounding $\hat{L}$ to two decimals (the interval becomes 1.76 to 1.78, which is hard to compare with 2b); ask for four. A “NaNs produced” warning is possible from some starting values and is harmless if the fit converges (`fit$convergence == 0`).

2b.

(Recommended: no AI.) Implement your profile from 1c and report the 95% profile interval for L . How does it compare with 2a?

```
L_grid <- seq(1.755, 1.790, by = 0.0001)
prof <- sapply(L_grid, function(L) {
  optimize(function(b) nll_ld50(c(L, b)), interval = c(5, 80))$objective
})
dprof <- prof - fit$value
prof_L <- range(L_grid[dprof <= qchisq(0.95, 1) / 2])
prof_L
```

```
## [1] 1.7640 1.7791
```

Answer. About 1.7640 to 1.7791. It is nearly identical to the quadratic interval from 2a.

Why. With about 480 beetles and $\hat{L}$ in the middle of the data, the profile is close to a parabola, so the quadratic approximation works well here.

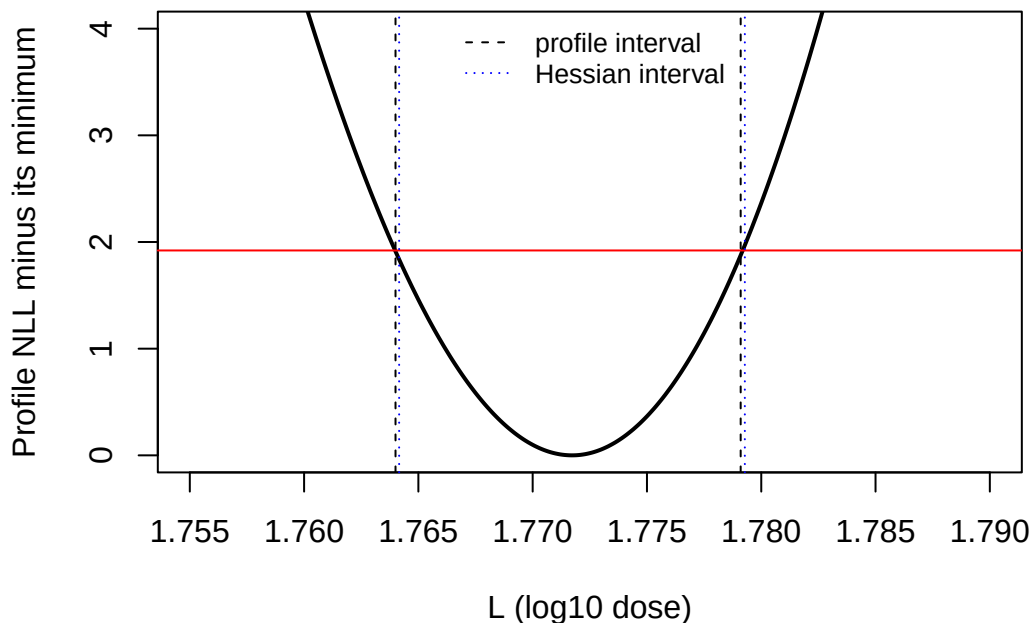
Watch for. Reusing `optim()` with a two-element starting vector inside the loop, so that L gets re-estimated instead of held fixed (an easy mistake to miss in AI-written code). Students may use `optim(method = "Brent")` or `optimize()` for the inner one-parameter fit; both are fine. Small differences in the fourth decimal come from grid spacing.

2c.

(AI optional/encouraged.) Plot your profile. Put L on the x-axis. On the y-axis, put the profile NLL at each value of L minus the smallest profile NLL, so the lowest point of the curve is at zero. Add a horizontal line at 1.92 and mark both intervals (from 2a and 2b). Then convert $\hat{L}$ and the profile interval to mg/L.

LD50 = $10^{\hat{L}}$: _____ mg/L, from _____ to _____ mg/L

```
par(mar = c(4, 4.5, 0.5, 0.5))
plot(L_grid, dprof, type = "l", lwd = 2, ylim = c(0, 4),
     xlab = "L (log10 dose)", ylab = "Profile NLL minus its minimum")
abline(h = qchisq(0.95, 1) / 2, col = "red")
abline(v = prof_L, lty = 2)
abline(v = wald_L, lty = 3, col = "blue")
legend("top", c("profile interval", "Hessian interval"), lty = c(2, 3),
     col = c("black", "blue"), bty = "n", cex = 0.8)
```



Answer. $LD_{50} = 10^{1.7717} \approx 59.1$ mg/L, from 58.1 to 60.1 mg/L.

Why. 10^L is increasing in L , so the endpoints convert directly.

Watch for. Figures without the zero-shift (the y-axis shows raw NLL around 18.7 or -148.8 and the 1.92 line is missing or misplaced). Figures where the two intervals can't be told apart are fine; they nearly coincide here, and saying so is a good answer. Converting with e^L .

2d.

Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?

Discussion points. No single correct answer. Useful responses name a specific step: the inner optimization in the profile loop, the sign on the Hessian, getting the plot's y-axis right. If an AI-written profile loop re-estimated L instead of holding it fixed, that's a good example to share (anonymously) with the class.

If you finish early.

(AI optional/encouraged.) The dose that kills 99% of beetles, L_{99} , satisfies $\log \frac{p}{1-p} = b(x - L_{99}) + \log \frac{0.99}{0.01}$. Profile L_{99} the same way and compare the width of its interval with the LD_{50} 's.

```
nll_ld99 <- function(par) {
  eta <- par[2] * (x - par[1]) + qlogis(0.99)
  -sum(dbinom(killed, size = n_exposed, prob = plogis(eta), log = TRUE))
}
fit99 <- optim(c(1.9, 34), nll_ld99)
L99_grid <- seq(1.86, 1.98, by = 0.0002)
prof99 <- sapply(L99_grid, function(L) {
  optimize(function(b) nll_ld99(c(L, b)), interval = c(5, 80))$objective
})
prof_L99 <- range(L99_grid[prof99 - fit99$value <= qchisq(0.95, 1) / 2])
c(fit99$par[1], prof_L99)
```

```
## [1] 1.905809 1.885400 1.930800
```

```
10^c(fit99$par[1], prof_L99)
```

[1] 80.50237 76.80686 85.27073

Answer. $\hat{L}_{99} \approx 1.906$ (80.5 mg/L), profile interval 1.885 to 1.931 (76.8 to 85.3 mg/L). On the log scale it is about 3 times as wide as the LD50's interval.

Why. See the reflection. Note that $\hat{L}_{99}$ is above the highest dose tested (1.8839), so this is an extrapolation.

Watch for. Forgetting the $\log(0.99/0.01)$ offset (that just refits the LD50). Reusing the LD50 grid, which is far too narrow.

Reflection

Suppose the pest manager wants a dose that kills 99% of beetles instead of 50%. Without computing anything, would you expect its interval to be wider or narrower than the LD50's? (Look at where the data sit in Act 1's figure.) The Hessian and profile intervals nearly agreed here. When would you expect them to disagree? Think back to the chytrid curve from Monday's WOD.

Sample answer. Wider. The doses with intermediate kill rates carry the most information about where the curve crosses 50%. Near 99%, a batch of 60 beetles can only show "all or nearly all died," and the LD99 lies above the highest dose tested, so the data say much less about where the curve reaches 99%. The LD99 also depends heavily on the slope b : a small change in b moves the 99% point a long way, while the 50% point barely moves. The Hessian and profile intervals disagree when the NLL curve isn't close to a parabola: small samples, estimates near a boundary (like prevalence near 0 in the chytrid WOD), or quantities estimated outside the range of the data.

Discussion points. Tie this back to the opening question: the LD50 is well determined by these data, but the dose a pest manager would actually use is not. For the LD99 the two methods still roughly agree here (the Hessian interval is 76.4 to 84.8 mg/L), but the profile is a little lopsided and shifted toward higher doses. Also accept answers that mention the curve being "flatter" at the edge of the data, which is the curvature idea from Monday's lecture.